

Executive Summary

This material covers the core principles of cellular biology, photosynthesis, cellular respiration, cell transport, DNA and protein synthesis, cell division, genetics, enzymes, homeostasis, and human body systems. It explains the structure and function of cells, energy transformations, genetic inheritance, and regulatory mechanisms essential for life.

KEY TAKEAWAYS

- The cell is the basic structural and functional unit of all living organisms, and all living things are composed of cells according to the cell theory.
- Photosynthesis transforms light energy into chemical energy in green plants, producing glucose and oxygen from carbon dioxide and water.
- Cellular respiration breaks down glucose to produce ATP, with the summary equation $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.
- The cell membrane controls substance movement through processes like diffusion, osmosis, active transport, and endocytosis.
- DNA carries genetic instructions, and protein synthesis involves transcription and translation guided by the genetic code.
- Mitosis produces two genetically identical daughter cells, while meiosis generates genetic diversity.
- Genes and alleles determine traits, with dominant and recessive inheritance patterns predicted using Punnett squares.
- Enzymes act as biological catalysts, with activity dependent on factors like temperature and substrate binding at the active site.
- Homeostasis maintains stable internal conditions through mechanisms like negative and positive feedback.
- Human body systems, including circulatory, respiratory, nervous, and endocrine systems, work together to sustain life.

Topics

TOPIC 1

Cellular Biology

Introduces the cell as the fundamental unit of life, describing its structure and key organelles.

Cell Theory

- The cell is defined as the basic structural and functional unit of all living organisms.
- All living things are composed of cells.
- New cells are produced from existing cells.

Cell Structure

- The nucleus is the membrane-bound organelle that contains the genetic material.
- The mitochondria is the organelle responsible for producing ATP through cellular respiration.

Organism Composition

- Every organism is composed of one or more cells.

TOPIC 2

Photosynthesis

Explains how green plants convert light energy into chemical energy stored in glucose.

Process Overview

- Photosynthesis is the process by which green plants transform light energy into chemical energy.
- Photosynthesis occurs in the chloroplast, which contains the green pigment chlorophyll.

Key Components

- Chlorophyll is the pigment that absorbs light energy.
- The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Reaction Stages

- The light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH.
- The Calvin cycle is the light-independent stage that fixes carbon dioxide into glucose.

TOPIC 3

Cellular Respiration

Describes the process by which cells convert glucose into ATP for energy.

Stages of Respiration

- Glycolysis occurs in the cytoplasm and is the first stage, during which glucose is broken down into two molecules of pyruvate.
- The Krebs cycle occurs in the mitochondrial matrix.
- The electron transport chain is the final stage and occurs on the inner mitochondrial membrane.

Anaerobic Processes

- Fermentation is the anaerobic process that regenerates NAD⁺ when oxygen is unavailable.

Summary Equation

- The summary equation of cellular respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.

TOPIC 4

Cell Transport

Explains how substances move across the selectively permeable cell membrane.

Passive Transport

- Diffusion is the passive movement of molecules from an area of high concentration to low concentration.
- Osmosis is defined as the diffusion of water across a selectively permeable membrane.

Active Transport

- Active transport is the movement of substances against their concentration gradient, requiring energy in the form of ATP.

Bulk Transport

- Endocytosis is the process by which the cell engulfs large particles by wrapping the membrane around them.

TOPIC 5

DNA and Protein Synthesis

Details how genetic information is stored, copied, and used to build proteins.

DNA Structure and Function

- DNA is defined as the molecule that carries the genetic instructions for the development and functioning of living organisms.

DNA Replication

- DNA replication is the process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.

Protein Synthesis Steps

- Transcription is the process by which the DNA sequence of a gene is copied into messenger RNA.
- Translation is defined as the process by which the ribosome builds a protein from the mRNA sequence.
- Each three-nucleotide codon on mRNA specifies one amino acid.
- The genetic code is the set of rules by which information encoded in genetic material is translated into proteins.

TOPIC 6

Cell Division

Describes the processes of mitosis and meiosis that distribute genetic material during cell division.

Mitosis Overview

- Mitosis is the process of nuclear division that produces two genetically identical daughter cells.

Cell Cycle

- The cell cycle consists of interphase, mitosis, and cytokinesis.
- Interphase is the stage during which the cell grows, copies its DNA, and prepares for division.

Mitosis Stages

- Prophase is the first stage of mitosis, during which chromosomes condense.
- Anaphase is the stage during which sister chromatids are pulled apart to opposite poles of the cell.

TOPIC 7

Genetics

Explains the principles of heredity, including genes, alleles, and inheritance patterns.

Genetic Terms

- A gene is defined as the basic unit of heredity, a segment of DNA that codes for a specific trait.
- An allele is an alternative form of a gene at the same locus.

Genotype and Phenotype

- The genotype is the genetic makeup of an organism.
- The phenotype is its observable characteristics.

Inheritance Patterns

- A dominant allele is expressed when only one copy is present.
- A recessive allele is only expressed when two copies are present.
- Punnett squares are used to predict the probability of offspring inheriting particular traits.

TOPIC 8

Enzymes

Describes how enzymes function as biological catalysts to speed up chemical reactions.

Enzyme Function

- An enzyme is defined as a protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.

Active Site and Binding

- The active site is the region of an enzyme where the substrate binds.
- The induced fit model describes how the enzyme changes shape to accommodate the substrate.

Enzyme Activity Factors

- Optimal temperature is the temperature at which an enzyme works fastest.
- Denaturation is defined as the loss of an enzyme's three-dimensional shape, which destroys its function.

Homeostasis

Explains how organisms maintain stable internal conditions despite external changes.

Homeostasis Definition

- Homeostasis is defined as the maintenance of a stable internal environment despite external changes.

Feedback Mechanisms

- Negative feedback is a mechanism that reverses a change, bringing the system back to its set point.
- Positive feedback amplifies a change and drives the system further from its set point.

Regulated Variables

- The normal human body temperature is maintained at approximately 37 degrees Celsius.
- Blood glucose concentration is regulated by the hormones insulin and glucagon.
- Insulin is secreted when blood glucose is high, promoting glucose uptake.

Human Systems

Describes the major organ systems in the human body and their functions.

Circulatory System

- The circulatory system transports oxygen, nutrients, and waste products throughout the body.
- The heart is the muscular organ that pumps blood through the circulatory system.

Respiratory System

- The respiratory system is responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.

Nervous System

- The nervous system coordinates rapid responses through electrical signals called nerve impulses.

Endocrine System

- The endocrine system regulates long-term processes through chemical messengers called hormones.

Terms & Definitions

Term	Definition
cell	The basic structural and functional unit of all living organisms. Every organism is composed of one or more cells.
cell theory	A fundamental principle of biology that states that all living things are composed of cells and that new cells are produced from existing cells.
nucleus	The membrane-bound organelle that contains the genetic material.
mitochondria	The organelle responsible for producing ATP through cellular respiration.
photosynthesis	The process by which green plants transform light energy into chemical energy.
chloroplast	The organelle where photosynthesis occurs and which contains the green pigment chlorophyll.
chlorophyll	The pigment that absorbs light energy during photosynthesis.
C₆H₁₂O₆	$6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
light-dependent reactions	Reactions that occur in the thylakoid membranes and produce ATP and NADPH.
Calvin cycle	The light-independent stage of photosynthesis that fixes carbon dioxide into glucose.
cellular respiration	The process that consists of three main stages: glycolysis, the Krebs cycle, and the electron transport chain.
glycolysis	The first stage of cellular respiration that occurs in the cytoplasm and breaks down glucose into two molecules of pyruvate.
Krebs cycle	The stage of cellular respiration that occurs in the mitochondrial matrix.
electron transport chain	The final stage of cellular respiration that occurs on the inner mitochondrial membrane.
fermentation	The anaerobic process that regenerates NAD ⁺ when oxygen is unavailable.
cell membrane	The selectively permeable barrier that controls what enters and leaves the cell.
diffusion	The passive movement of molecules from an area of high concentration to low concentration.
osmosis	The diffusion of water across a selectively permeable membrane.
active transport	The movement of substances against their concentration gradient, requiring energy in the form of ATP.
endocytosis	The process by which the cell engulfs large particles by wrapping the membrane around them.
DNA	The molecule that carries the genetic instructions for the development

Term	Definition
	and functioning of living organisms.
DNA replication	The process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.
transcription	The process by which the DNA sequence of a gene is copied into messenger RNA.
translation	The process by which the ribosome builds a protein from the mRNA sequence.
genetic code	The set of rules by which information encoded in genetic material is translated into proteins.
mitosis	The process of nuclear division that produces two genetically identical daughter cells.
cell cycle	Consists of interphase, mitosis, and cytokinesis.
interphase	The stage during which the cell grows, copies its DNA, and prepares for division.
prophase	The first stage of mitosis, during which chromosomes condense.
anaphase	The stage during which sister chromatids are pulled apart to opposite poles of the cell.
gene	The basic unit of heredity, a segment of DNA that codes for a specific trait.
allele	An alternative form of a gene at the same locus.
genotype	The genetic makeup of an organism.
phenotype	The observable characteristics of an organism.
dominant allele	An allele that is expressed when only one copy is present.
recessive allele	An allele that is only expressed when two copies are present.
Punnett square	A tool used to predict the probability of offspring inheriting particular traits.
enzyme	A protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.
active site	The region of an enzyme where the substrate binds.
induced fit model	Describes how the enzyme changes shape to accommodate the substrate.
optimal temperature	The temperature at which an enzyme works fastest.
denaturation	The loss of an enzyme's three-dimensional shape, which destroys its function.

Term	Definition
homeostasis	The maintenance of a stable internal environment despite external changes.
negative feedback	A mechanism that reverses a change, bringing the system back to its set point.
positive feedback	A mechanism that amplifies a change and drives the system further from its set point.
insulin	A hormone secreted when blood glucose is high, promoting glucose uptake.
glucagon	A hormone that regulates blood glucose concentration.
circulatory system	The system that transports oxygen, nutrients, and waste products throughout the body.
heart	The muscular organ that pumps blood through the circulatory system.
respiratory system	The system responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.
nervous system	The system that coordinates rapid responses through electrical signals called nerve impulses.
endocrine system	The system that regulates long-term processes through chemical messengers called hormones.
ATP	The molecule produced by mitochondria through cellular respiration, providing energy for cellular processes.

Key Facts & Formulas

Formula / Fact	Context
respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.	respiration is $C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$.
$C_6H_{12}O_6$	produce glucose and release oxygen. The summary equation of photosynthesis is $6CO_2 + 6H_2O = C_6H_{12}O_6 + 6O_2$. The light-dependent reactions occur in the thylakoid membranes and produce ATP and